

Fusion-JEPA Text-to-Speech

Week 7 Progress Report

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Audio Generation Progress

- **Significant Improvements:** We achieved noticeably better generation results over the last two weeks of continuous training.
- **Audio Output:** The SpatialDiT diffuser is finally outputting coherent audio waveforms.
- **Acoustic Quality:** The generated samples sound distinctly like English speech, though the prosody and clarity are not quite perfect yet.

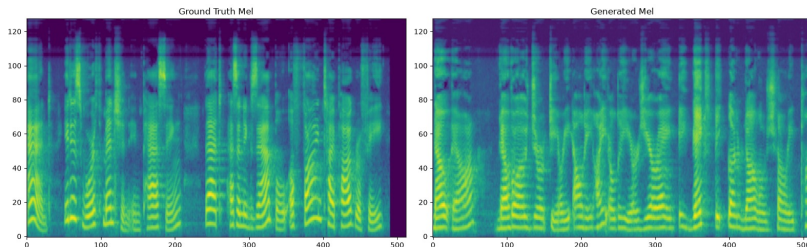


Figure: Recent audio generation output showing strong structural foundations.

Architectural Enhancements

To achieve these results, we implemented several major changes to the model architecture:

- **Single-Stream Architecture:** We completely removed the Cross-Attention mechanism. Phoneme text embeddings are now concatenated directly to the visual tokens, forcing the model to process them in a unified single stream.
- **Removed Mean-Pooling:** We eliminated the mean-pooling layers to preserve the sequential alignment of phonemes, fixing the temporal destruction bottleneck.
- **Cosine Annealing:** Replaced static learning rates with a Cosine Annealing LR Scheduler to ensure smooth decay and convergence.
- **Targeted Freezing:** We introduced a training mechanism to completely freeze the SpatialDiT diffuser, allowing the JEPA backbone to learn mapping exclusively.

Current Training Status

- **JEPA Backbone Training:** As we speak, the model is actively training the JEPA backbone while the diffuser remains strictly frozen.
- **Cloud Sync:** The training pipeline is fully automated and is uploading checkpoints directly to Hugging Face in real-time.

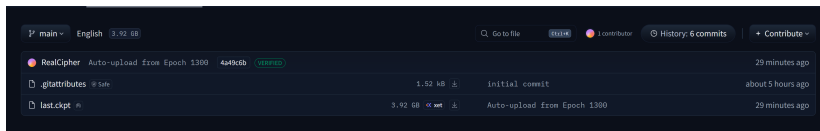


Figure: Live training logs and Hugging Face checkpoint uploads.

Future Work & Next Steps

- **Diffuser Fine-Tuning:** Once the JEPA backbone has fully converged and learned robust text-to-spectrogram mappings, we will unfreeze the SpatialDiT diffuser. The diffuser will then be fine-tuned on the significantly better structural inputs provided by the JEPA.
- **Arabic Model Training:** Following the successful optimization and training of the English model pipeline, we will initiate training for the Arabic model utilizing the exact same single-stream architecture.